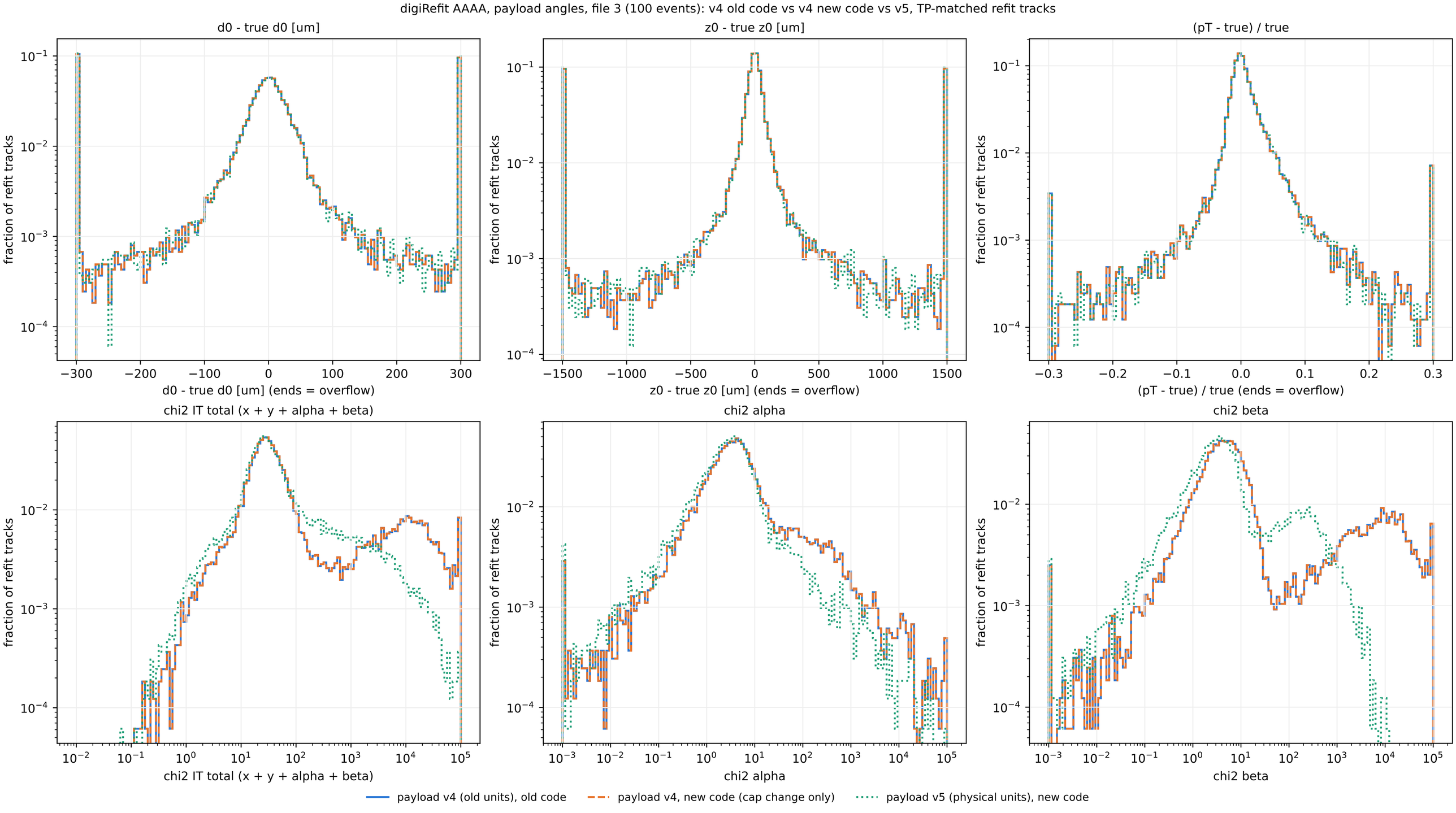
digiRefit AAAA, payload angles, file 3 (100 events): v4 old code vs v4 new code vs v5, TP-matched refit tracks
d0 - true d0 [um]
z0 - true z0 [um]
(pT - true) / true
chi2 IT total (x + y + alpha + beta)
chi2 alpha
chi2 beta
fraction of refit tracks
d0 - true d0 [um] (ends = overflow)
z0 - true z0 [um] (ends = overflow)
(pT - true) / true (ends = overflow)
chi2 IT total (x + y + alpha + beta)
chi2 alpha
chi2 beta
payload v4 (old units), old code
payload v4, new code (cap change only)
payload v5 (physical units), new code

Refit (TP-matched refit tracks): residual q68 half-widths and medians vs the TP; chi2 summaries (1)

run	quantity	tp_pt	n	q68_halfwidth	median	mean	q90	q99	frac_gt_100
payload v4 (old units), old code	d0 [um]	[0, inf)	16303	58.16	0.03152	nan	nan	nan	nan
payload v4 (old units), old code	d0 [um]	[2, 5)	13309	55.35	0.03152	nan	nan	nan	nan
payload v4 (old units), old code	d0 [um]	[5, inf)	1666	24.89	0.5178	nan	nan	nan	nan
payload v4 (old units), old code	z0 [um]	[0, inf)	16303	150	0.4888	nan	nan	nan	nan
payload v4 (old units), old code	z0 [um]	[2, 5)	13309	136.8	0.4816	nan	nan	nan	nan
payload v4 (old units), old code	z0 [um]	[5, inf)	1666	54.47	0.4601	nan	nan	nan	nan
payload v4 (old units), old code	pT (reco - true) / true	[0, inf)	16303	0.01949	0.001336	nan	nan	nan	nan
payload v4 (old units), old code	pT (reco - true) / true	[2, 5)	13309	0.01818	0.0005588	nan	nan	nan	nan
payload v4 (old units), old code	pT (reco - true) / true	[5, inf)	1666	0.01785	0.001834	nan	nan	nan	nan
payload v4 (old units), old code	chi2 IT total	all	16303	nan	36.86	4690	1.103e+04	8.047e+04	0.284
payload v4 (old units), old code	chi2 alpha	all	16303	nan	3.815	291.2	70.89	3285	0.08643
payload v4 (old units), old code	chi2 beta	all	16303	nan	5.811	3780	8250	7.145e+04	0.2249
payload v4, new code (cap change only)	d0 [um]	[0, inf)	16303	58.16	0.03152	nan	nan	nan	nan
payload v4, new code (cap change only)	d0 [um]	[2, 5)	13309	55.35	0.03152	nan	nan	nan	nan
payload v4, new code (cap change only)	d0 [um]	[5, inf)	1666	24.89	0.5178	nan	nan	nan	nan
payload v4, new code (cap change only)	z0 [um]	[0, inf)	16303	150	0.4888	nan	nan	nan	nan
payload v4, new code (cap change only)	z0 [um]	[2, 5)	13309	136.8	0.4816	nan	nan	nan	nan
payload v4, new code (cap change only)	z0 [um]	[5, inf)	1666	54.47	0.4601	nan	nan	nan	nan
payload v4, new code (cap change only)	pT (reco - true) / true	[0, inf)	16303	0.01949	0.001336	nan	nan	nan	nan
payload v4, new code (cap change only)	pT (reco - true) / true	[2, 5)	13309	0.01818	0.0005588	nan	nan	nan	nan
payload v4, new code (cap change only)	pT (reco - true) / true	[5, inf)	1666	0.01785	0.001834	nan	nan	nan	nan
payload v4, new code (cap change only)	chi2 IT total	all	16303	nan	36.86	4690	1.103e+04	8.047e+04	0.284
payload v4, new code (cap change only)	chi2 alpha	all	16303	nan	3.815	291.2	70.89	3285	0.08643
payload v4, new code (cap change only)	chi2 beta	all	16303	nan	5.811	3780	8250	7.145e+04	0.2249
payload v5 (physical units), new code	d0 [um]	[0, inf)	16303	60.22	0.008241	nan	nan	nan	nan
payload v5 (physical units), new code	d0 [um]	[2, 5)	13309	56.94	0.03327	nan	nan	nan	nan
payload v5 (physical units), new code	d0 [um]	[5, inf)	1666	25.25	0.3903	nan	nan	nan	nan
payload v5 (physical units), new code	z0 [um]	[0, inf)	16303	157.8	0.2301	nan	nan	nan	nan
payload v5 (physical units), new code	z0 [um]	[2, 5)	13309	142.5	0.2718	nan	nan	nan	nan
payload v5 (physical units), new code	z0 [um]	[5, inf)	1666	57.95	0.2813	nan	nan	nan	nan

Refit (TP-matched refit tracks): residual q68 half-widths and medians vs the TP; chi2 summaries (2)

run	quantity	tp_pt	n	q68_halfwidth	median	mean	q90	q99	frac_gt_100
payload v5 (physical units), new code	pT (reco - true) / true	[0, inf)	16303	0.01959	0.001407	nan	nan	nan	nan
payload v5 (physical units), new code	pT (reco - true) / true	[2, 5)	13309	0.01829	0.0005426	nan	nan	nan	nan
payload v5 (physical units), new code	pT (reco - true) / true	[5, inf)	1666	0.01858	0.001878	nan	nan	nan	nan
payload v5 (physical units), new code	chi2 IT total	all	16303	nan	32.15	754.8	830.3	1.648e+04	0.2325
payload v5 (physical units), new code	chi2 alpha	all	16303	nan	2.99	85.25	14.95	1017	0.03349
payload v5 (physical units), new code	chi2 beta	all	16303	nan	3.668	72.46	152.4	1300	0.1251

Accepted refit hits: selected-hit purity (selHitClass 0 = own TP, 1 = other TP, 2 = no TP), has* rates, and the refit |cot| <= 12 cap (violations must be 0)

run	accepted_hits	purity_own_TP	frac_other_TP	frac_no_TP	hasAlpha_frac	hasBeta_frac	cap_rule_violations	accepted_with_abs_cot_gt12	refit_tracks
payload v4 (old units), old code	46548	0.8601	0.1263	0.01364	0.9993	0.9987	0	68	16303
payload v4, new code (cap change only)	46548	0.8601	0.1263	0.01364	0.9993	0.9987	0	68	16303
payload v5 (physical units), new code	46518	0.8548	0.1296	0.01554	0.9992	0.9987	0	75	16303